퀴 즈 (6)회

단원	내용	학과	학번	성명
14.1-14.3	직사각형과 일반영역 위의 이중적분/극좌표에서의 이중적분	전전컴	2022440032	김예영

1. $R=[0,\,1]\times[0,\,2]$ 일 때, $\iint_R x^3\sin(x^2y)\,dA$의 값을 구하여라. (2점)

$\int_0^1\int_0^2 x^3\sin(x^2y)\,dy\,dx$

$[-x\cos(x^2y)]_0^2 = -x\cos(2x^2)+x$

$\int_0^1 (x-x\cos(2x^2))\,dx$

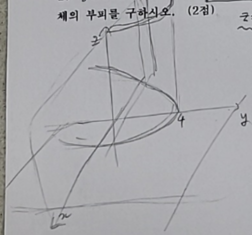
$=\left[\frac{1}{2}x^2-\frac{1}{4}\sin(2x^2)\right]_0^1$

$=\frac{1}{2}-\frac{1}{4}\sin 2$

$\boxed{\frac{1}{2}-\frac{1}{4}\sin 2}$

2. $y=4-x^2$, $z=0$, $y=0$, $\boxed{y+z=6}$에 의해 유계된 입체의 부피를 구하시오. (2점)

$z=6-y$



$\left[6y-\frac{1}{2}y^2\right]_0^{4-x^2}$

$24-6x^2-\frac{1}{2}(16-8x^2+x^4)$

$f(x,y)=6-y$

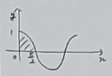
$D=\{(x,y)\mid -2\le x\le 2,\ 0\le y\le 4-x^2\}$

$\int_{-2}^{2}\int_0^{4-x^2}(6-y)\,dy\,dx$

$=\int_{-2}^{2}\left(-\frac{1}{2}x^4-2x^2+16\right)dx = 2\left[-\frac{1}{10}x^5-\frac{2}{3}x^3+16x\right]_0^2$

$=\frac{704}{15}$

3. 적분 $\int_0^1\int_0^{\cos^{-1}y}\sin^2x\sqrt{1+2\sin^3x}\,dx\,dy$의 값을 구하여라. (2점) $\cos^{-1}y=x$ $y=\cos x$



$D=\{(x,y)\mid 0\le x\le\cos^{-1}y,\ 0\le y\le 1\}$

$=\{(x,y)\mid 0\le x\le\frac{\pi}{2},\ 0\le y\le\cos x\}$

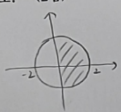
$\int_0^{\frac{\pi}{2}}\int_0^{\cos x}\sin^2x\sqrt{1+2\sin^3x}\,dy\,dx$

$=\int_0^{\frac{\pi}{2}}\cos x\sin^2x\sqrt{1+2\sin^3x}\,dx$

$=\left[\frac{2}{3}\times\frac{1}{6}(1+2\sin^3x)^{\frac{3}{2}}\right]_0^{\frac{\pi}{2}}$

$=\boxed{\frac{1}{9}(3\sqrt{3}-1)}$

4. 이중적분 $\int_0^2\int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}}e^{-x^2-y^2}\,dy\,dx$의 값을 구하시오. (2점)



$x^2+y^2=r^2$

$D=\{(r,\theta)\mid 0\le r\le 2,\ \frac{-\pi}{2}\le\theta\le\frac{\pi}{2}\}$

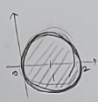
$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\int_0^2 e^{-r^2}r\,dr\,d\theta$

$=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}-\frac{1}{2}(e^{-4}-1)\,d\theta$

$=\boxed{-\frac{\pi}{2}(e^{-4}-1)}$

5. 이중적분 $\int_0^2\int_0^{\sqrt{2x-x^2}}\sqrt{x^2+y^2}\,dy\,dx$의 값을 구하시오. (2점)

$y^2=2x-x^2$ $(x-1)^2+y^2=1$



$\boxed{\frac{4}{3}\pi}$